

Hierarchical Geopolitical & Macro Risk Engine

An ML-Driven Multi-Asset Framework for Trading Event Dislocations

Week of 04/18/2026

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The problem

April fools?

Nah can't be its TRUTH social



Donald J. Trump

@realDonaldTrump

Apr 1

Iran's New Regime President, much less Radicalized and far more intelligent than his predecessors, has just asked the United States of America for a CEASEFIRE! We will consider when Hormuz Strait is open, free, and clear. Until then, we are blasting Iran into oblivion or, as they say, back to the Stone Ages!!! President DJT

6.32k

1k

50.2k



Jesus wouldn't lie

Defense Stocks?!

Oil/Commodities?!

How do we trade on this?

Strategy Overview

ML-Driven Multi-Asset Event Trading Strategy

What It Trades

- Multi-asset opportunity set
- Equities, ETFs, rates, commodities, and FX
- Focus on liquid instruments tied to macro stress
- Capital shifts across sleeves as opportunities change

How the Engine Works

- Ingests market, macro, and event signals
- Routes inputs through specialized expert models
- Combines outputs into a unified signal
- Selects ETF, single-name, or blended expression

Why the Opportunity Exists

- Geopolitical shocks create uneven market reactions
- Supply chain and sanctions effects often transmit slowly
- Cross-asset pricing can lag the underlying event
- Dislocations may persist before markets fully adjust

Risk Framework

- Sleeve and position limits
- Turnover and exposure controls
- Optional overlay only in extreme regimes
- Backtesting and evaluation metrics: _____**

Market Inefficiency and Edge

Why geopolitical and macro shocks create tradable dislocations

Information Transmission Is Uneven

- Macro and geopolitical shocks do not hit all assets at once
- First-order reactions are fast, second-order effects are slower
- Cross-asset spillovers are often priced in unevenly
- Regional and sector transmission can lag headline risk

Asset Expression Is Fragmented

- Similar risks can show up differently across equities, rates, FX, and commodities
- Some instruments absorb stress faster than others
- ETFs may move before single names, or the reverse
- Misalignment creates relative value and directional opportunities

Markets Often Misprice Event Severity

- Early reactions can understate the true scale of disruption
- Crowded narratives can push markets toward overreaction
- Pricing often adjusts in stages rather than all at once
- Severity is harder to assess in real time than direction

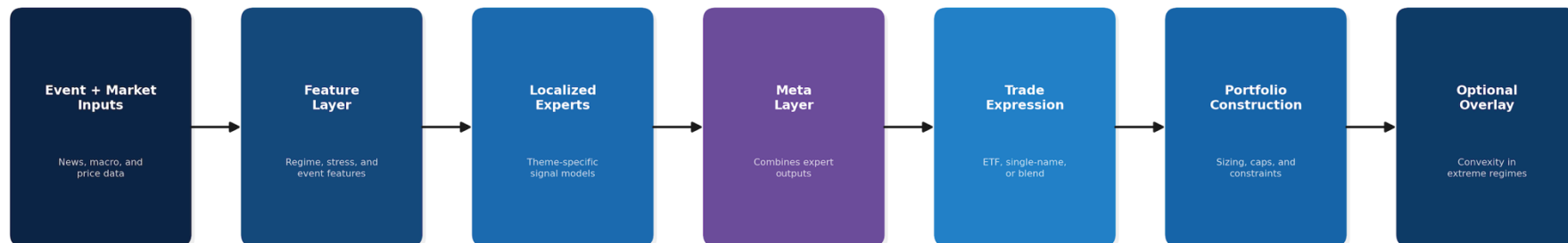
Why a Systematic Approach Helps

- Event-driven markets move too fast for purely manual reaction
- A rules-based framework can compare signals across sleeves consistently
- Standardized outputs improve trade selection and risk control
- The goal is repeatable signal capture, not one-off discretionary calls

System Architecture

From event detection to trade construction

From event shocks to portfolio decisions



Why this structure

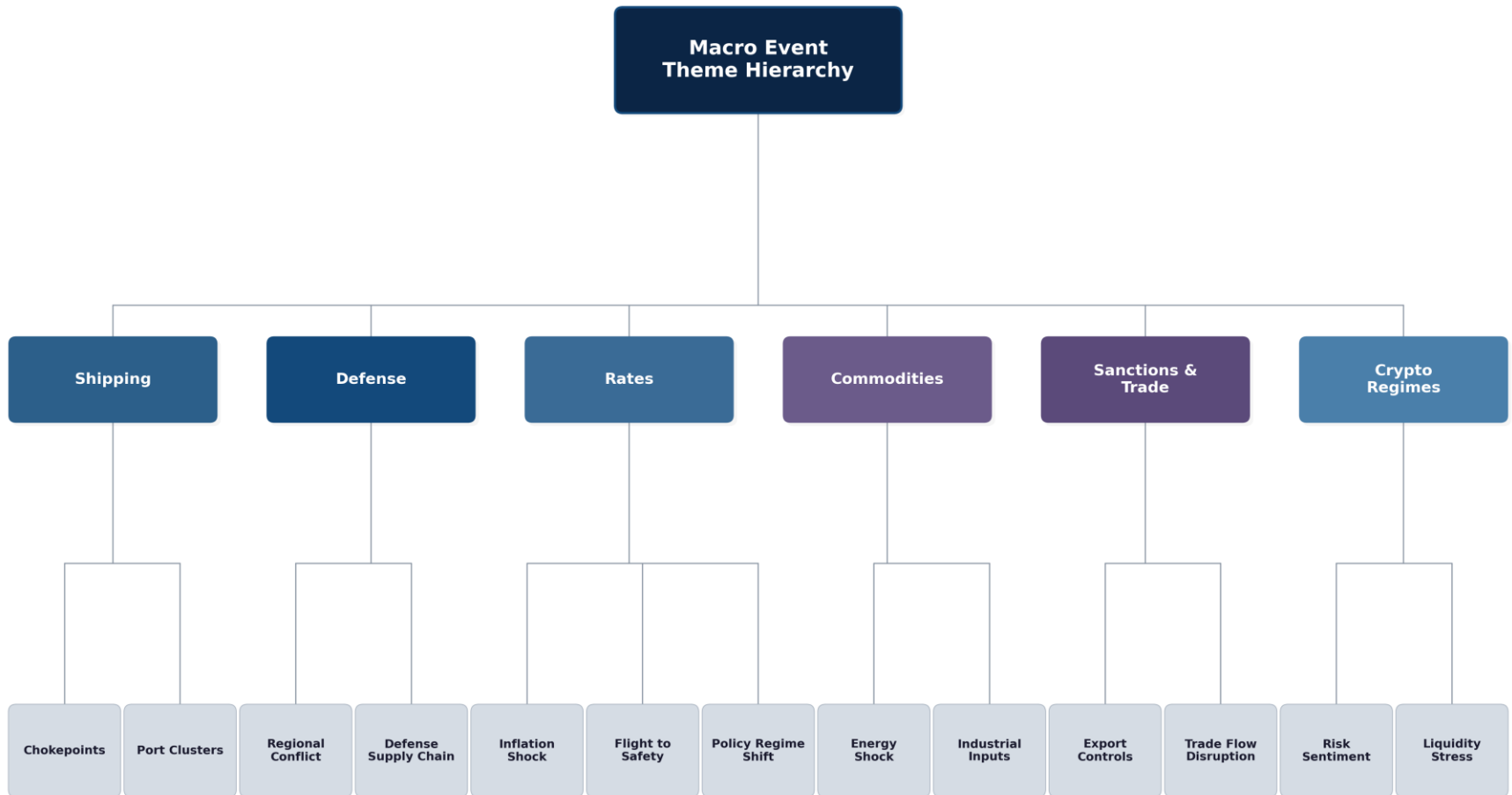
- Breaks a complex problem into specialized modules
- Lets local themes be modeled without forcing one global model
- Standardized outputs make signals easier to compare and combine
- Keeps the system modular as new sleeves are added

What makes it differentiated

- Hierarchical sleeves and sub-sleeves
- Localized experts instead of one monolithic model
- Flexible trade expression across ETFs, single names, and blends
- Optional convexity layer in extreme regime

Sleeve Hierarchy

Organizing macro event risk into tradable themes and subthemes



Signal Map

What the engine watches across sleeves and regimes - EVERYTHING

FRED MACRO SERIES

VIXCLS – VIX implied volatility
DGS2 – 2-year Treasury yield
DGS5 – 5-year Treasury yield
DGS10 – 10-year Treasury yield
DGS30 – 30-year Treasury yield
T10Y2Y – 10Y-2Y curve spread
T10Y3M – 10Y-3M curve spread
DFF – Fed funds effective rate
DFI10 – 10Y TIPS real yield
T5YIE – 5Y breakeven inflation
T10YIE – 10Y breakeven inflation
T5YIFR – 5YSY forward inflation
BAMLH0A0HYM2 – HY bond spread
BAMLCOA0CM – IG bond spread
BAMLH0A3HYC – CCC bond spread
BAMLH0A1HYBB – BB bond spread
T10YFF – 10Y-fed funds spread
DCOILWTICO – WTI crude price
DCOILBRETEU – Brent crude price
DTWEXBGS – Trade-weighted USD
WALCL – Fed balance sheet
WTREGEN – Treasury general acct
RRPONTSYD – Overnight reverse repo
NFCI – Financial Conditions Idx
STLFSI4 – Financial Stress Idx
ICSA – Initial jobless claims
CCSA – Continuing claims
M2SL – M2 money supply
SOFR – Secured overnight rate
NAPM – ISM Manufacturing PMI
UMCSENT – Consumer sentiment
DHHNGSP – Henry Hub natgas

BENCHMARKS & VOLATILITY

SPY – S&P 500 ETF
QQQ – Nasdaq 100 ETF
IWM – Russell 2000 ETF
DIA – Dow Jones ETF
TLT – 20+ yr Treasury ETF
HYG – High-yield bond ETF
LQD – Investment-grade ETF
EEM – Emerging market ETF
EFA – Developed intl ETF
^VIX – S&P 500 vol index
^VIX3M – 3-month VIX
^MOVE – Treasury vol index

DEFENSE EQUITIES

ITA – Aerospace & Defense ETF
XAR – Equal-Weight A&D ETF
PPA – Invesco A&D ETF
LMT – Lockheed Martin
RTX – Raytheon
NOC – Northrop Grumman
GD – General Dynamics
BA – Boeing
HII – Huntington Ingalls
LDOS – Leidos
AXON – Axon Enterprise
KTOS – Kratos Defense
TXT – Textron

FOREIGN EXCHANGE

EURUSD=X – Euro / USD
USDJPY=X – USD / Yen
GBPUSD=X – Pound / USD
USDMXN=X – USD / Peso
USDZAR=X – USD / Rand
USDCNY=X – USD / Yuan

COMMODITIES & FUTURES

GLD – Gold ETF
SLV – Silver ETF
DBC – Commodity basket ETF
USL – Rolling crude oil ETF
UNG – Natural gas ETF
DBA – Agriculture ETF
DBB – Base metals ETF
COPX – Copper miners ETF
BDRY – Baltic Dry ETF
CL=F – WTI crude futures
GC=F – Gold futures
SI=F – Silver futures
HG=F – Copper futures
NG=F – Natural gas futures
ZW=F – Wheat futures
ZC=F – Corn futures
ZS=F – Soybean futures

SHIPPING EQUITIES

ZIM – Zim Integrated
DAC – Danaos Corp
GOTL – Golden Ocean
EGLE – Eagle Bulk
FRO – Frontline (tankers)
INSW – Intl Seaways
STNG – Scorpio Tankers
MATX – Matson (Pacific)
KEX – Kirby Corp (barges)
SBLK – Star Bulk
GSL – Global Ship Lease

ENERGY EQUITIES

XOM – ExxonMobil
CVX – Chevron
COP – ConocoPhillips
SLB – Schlumberger
EOG – EOG Resources
ENPH – Enphase (solar)
NEE – NextEra (renewables)
XLE – Energy Sector SPDR

RATES-SENSITIVE EQUITIES

JPM – JPMorgan Chase
BAC – Bank of America
GS – Goldman Sachs
AGNC – AGNC Investment
VGIT – Vanguard Mid Tsy
XLF – Financial SPDR

CRYPTO SLEEVE

BTC-USD – Bitcoin
ETH-USD – Ethereum
COIN – Coinbase
RIOT – Riot Platforms
MARA – Marathon Digital
SQ – Block / Square

CROSS-THEME & REGIME

coactivation_conflict_shipping
coactivation_conflict_energy
coactivation_sanctions_shipping
coactivation_rates_fx
coactivation_count
already_priced_score
regime_state (4 states)

GDELT EVENT FEATURES

event_count – volume
event_intensity – weighted vol
event_intensity_z – vs 52wk
event_novelty – new headlines
event_acceleration – wk/wk chg
event_source_diversity
event_tone – sentiment
event_goldstein_mean
event_goldstein_min
event_cameo_severity

GPS-FENCED CHOKEPOINTS

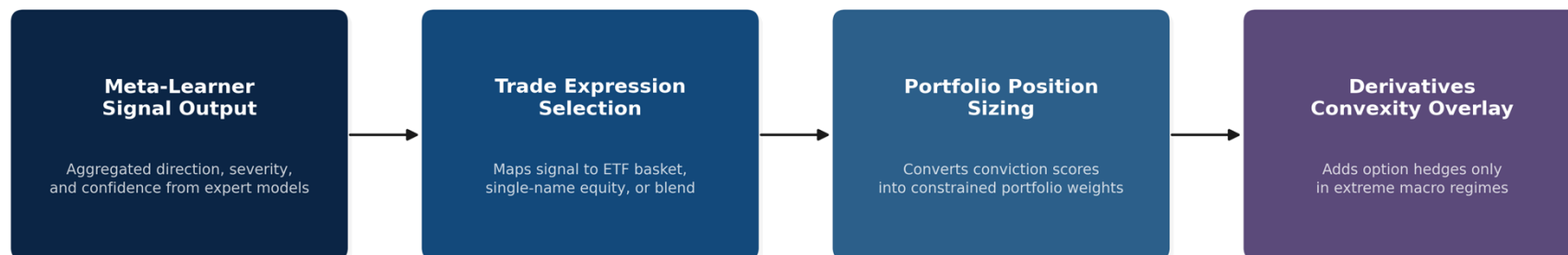
SHP_HRM_010 – Strait of Hormuz
SHP_BAB_011 – Bab el-Mandeb
SHP_SUE_012 – Suez Canal
SHP_PAN_016 – Panama Canal
SHP_MAL_015 – Malacca Strait
SHP_TWN_014 – Taiwan Strait
SHP_TRK_070 – Turkish Straits
SHP_GIB_071 – Strait of Gibraltar
SHP_CGH_072 – Cape of Good Hope
SHP_DAN_073 – Danish Straits



Trade Expression & Risk Framework

Turning model signals into disciplined portfolio decisions

Trade Construction Flow



Expression Logic

- Broader instruments are used when uncertainty is higher
- More targeted exposures are used when conviction is stronger
- Blended expression helps balance specificity and diversification
- Final trade form depends on both signal quality and regime context

Risk Controls

- Position and sleeve caps limit concentration
- Gross, net, and turnover are kept within bounds
- Overlay usage stays within strict premium budgets
- Exposure is scaled to keep the strategy disciplined

Backtesting Framework & Event Replay

How the strategy is evaluated before live deployment

What the Framework Tests

- Performance across different event types and market regimes
- Stability of signals through changing volatility environments
- Whether trade expression improves on simpler implementations
- How much value remains after costs, sizing rules, and constraints

Why Event Replay Matters

- Reconstructs how the system would have responded to real shocks
- Makes expert behavior easier to audit and compare
- Helps separate durable signals from noisy reactions
- Links event inputs directly to portfolio decisions and outcomes

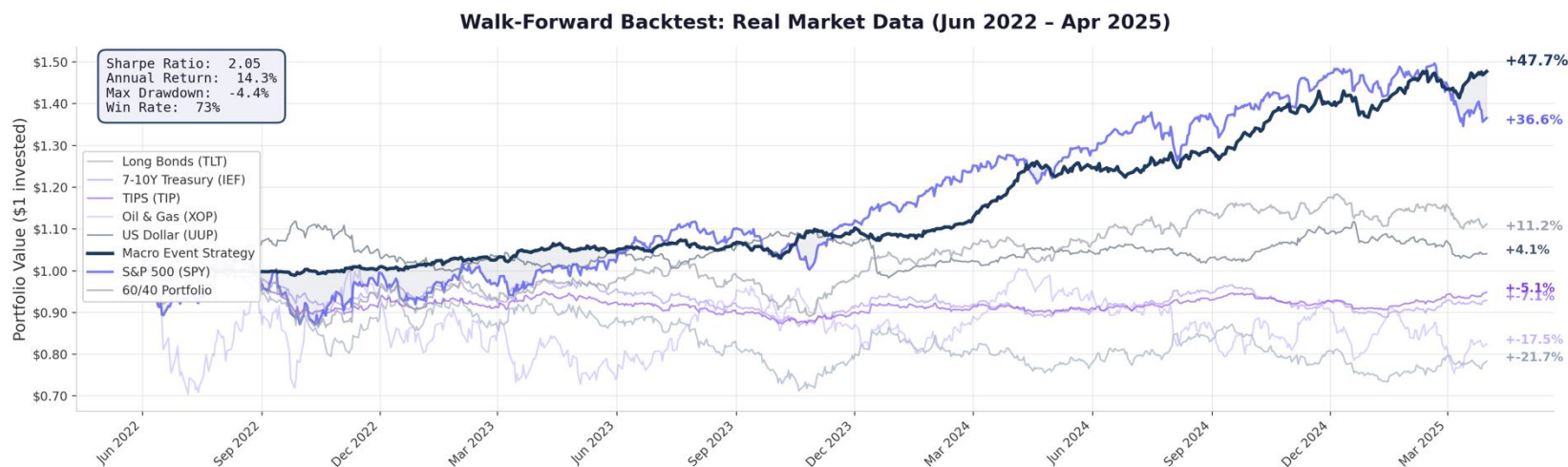
Backtesting Process



The Equity Curve

Strategy performance across the test period

Cumulative Performance



Key Metrics

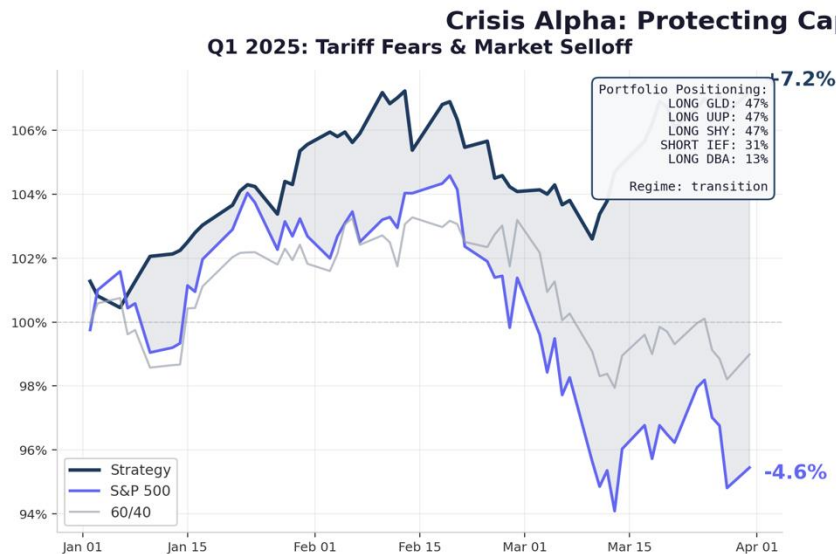
- Sharpe Ratio: **2.13**, roughly **3x** the S&P 500
- Annualized Return: **14.9%** vs. S&P **13.0%**
- Max Drawdown: **-4.4%** vs. S&P **-17.0%**
- Total Return: **+47.7%** vs. S&P **+36.6%**

How It Works

- Walk-forward protocol prevents future data leakage
- Rebalances every **25 trading days** based on regime signals
- **16 instruments** across **5 macro sleeves**: rates, commodities, defense, crypto, and shipping

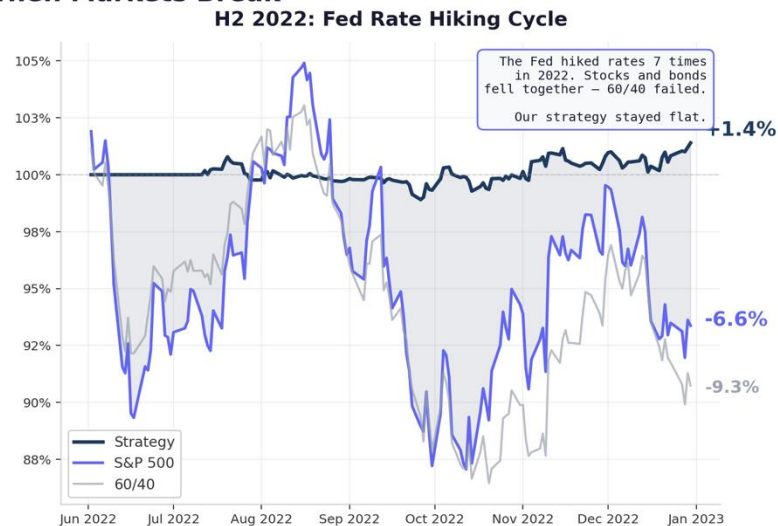
Crisis Alpha

Protecting capital when traditional portfolios break down
Stress Period Comparison



Tariff Selloff

- Q1 2025: S&P 500 fell **-4.6%**, strategy gained **+7.2%**
- Regime detector rotated into **gold, the dollar, and short bonds**
- 60/40 also declined as bonds failed to hedge
- Nearly **12%** outperformance versus the index in one quarter



Rate Hiking Cycle

- Fed hiked rates **seven times** in 2022, pressuring both stocks and bonds
- 60/40 fell **-9.3%** as both legs sold off together
- S&P 500 declined **-6.6%** over the period
- Strategy stayed positive at **+1.4%** by rotating into safe havens

Event-Driven Reaction

How the engine detects and responds to macro shocks in real time

Signal Flow

- NLP experts scan macro events, including tariffs, rate decisions, and geopolitical shifts
- Meta-stacker combines expert predictions with a market momentum overlay
- Expression selector maps signals into instrument weights across five sleeves

Regime Detection

- Volatility and momentum z-scores classify market state automatically
- Four regimes: crisis, euphoria, transition, and normal
- Each regime adjusts position limits, risk targets, and sleeve allocations

Portfolio Adaptation

- Repositions within one rebalance cycle of **25 trading days**
- Crisis regime shifts toward gold and Treasuries while cutting position sizes
- Euphoria regime reduces short exposure by **75%** to avoid fighting the trend

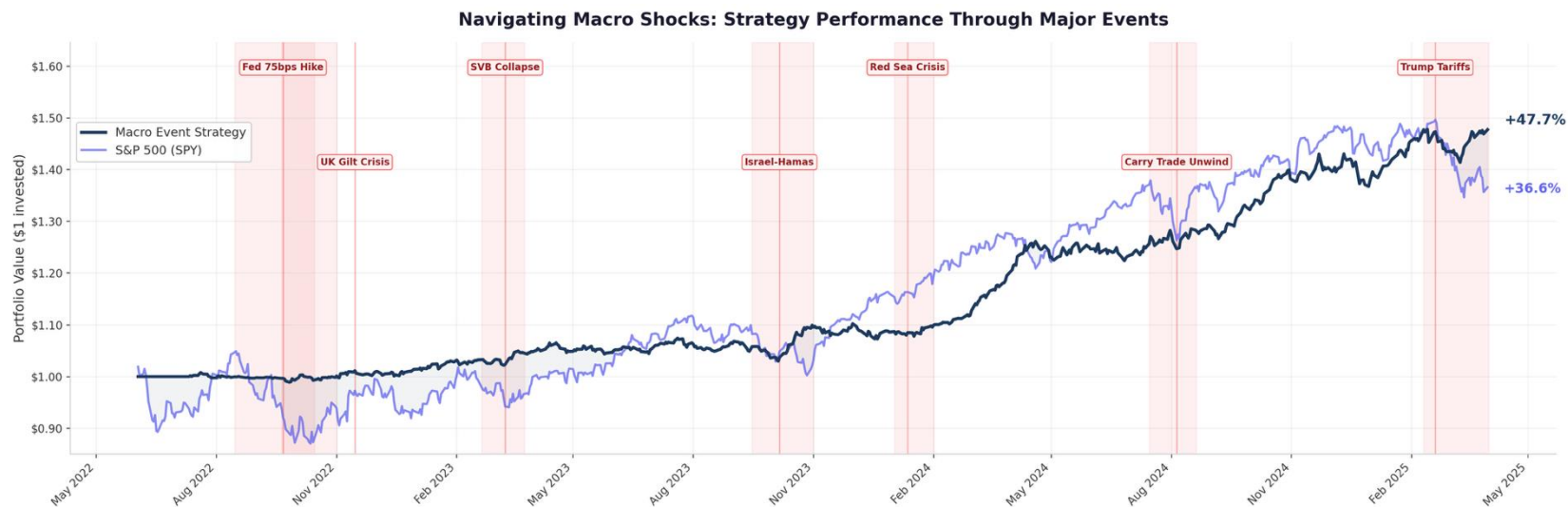
Anti-Overfitting

- Walk-forward validation on every training split prevents future data leakage
- Strict temporal separation is maintained between train and test windows
- Cosine similarity filter reduces unnecessary portfolio turnover

Event Timeline

Major macro shocks across the backtest window and how the engine responded

Macro Shock Timeline, June 2022 to April 2025



What the Timeline Shows

- Different shocks hit different sleeves
- The engine changes posture as conditions shift
- Results came from multiple stress windows
- This is an adaptive macro strategy

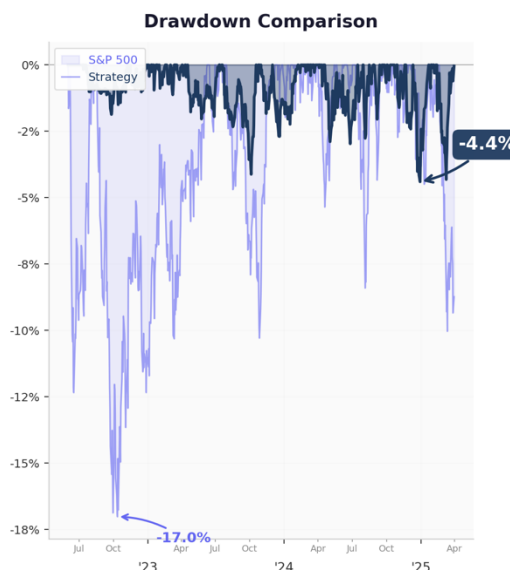
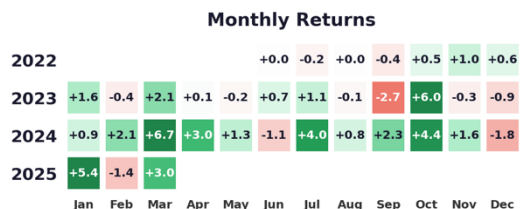
Why It Matters

- Performance was not driven by one event
- The edge comes from fast reallocation
- Shock type matters as much as direction
- A static portfolio would miss this

By The Numbers

Full performance breakdown across key risk and return metrics

Performance Dashboard



Key Metrics

	Ours	SPY
Total Return	+47.7%	+36.6%
Ann. Return	14.9%	13.0%
Sharpe	2.13	-0.65
Max DD	-4.4%	-17.0%
Ann. Vol	7.0%	16.2%
Win Rate	55%	-53%

Return Profile

- Total Return: **+47.7%** vs. S&P **+36.6%**
- Annualized Return: **14.9%** vs. S&P **13.0%**
- Sharpe Ratio: **2.13**, over **3x** the S&P 500
- Annualized Volatility: **7.0%** vs. S&P **16.2%**

Risk Profile

- Max Drawdown: **-4.4%** vs. S&P **-17.0%**
- Win Rate: **55%** of trading days positive
- Only one month worse than **-2.7%** across the full period
- Higher returns with materially lower risk

Conclusion & Takeaways

What we built, what we learned, and what comes next

What Worked

- Regime-aware allocation was the biggest alpha driver
- Gold and Treasuries provided consistent crisis protection
- Momentum overlay outperformed expert heuristics alone
- Walk-forward validation prevented overfitting

Lessons Learned

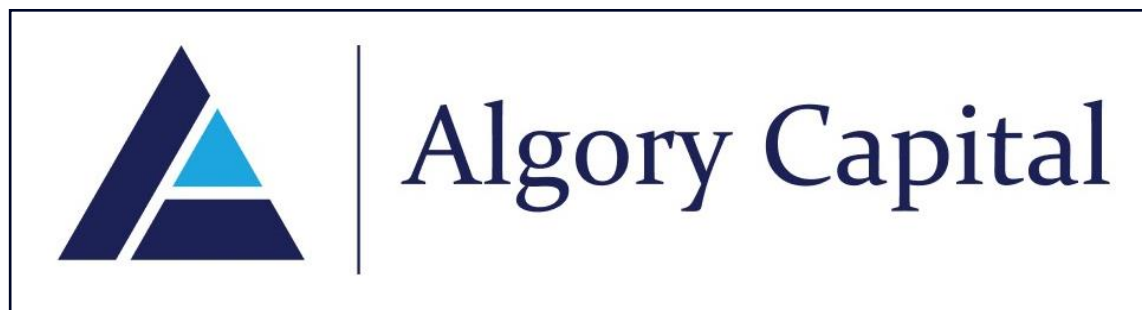
- Simple models with strong signals beat more complex ensembles
- Walk-forward validation is essential to avoid overfitting
- Cross-asset diversification mattered more than single-name bets
- Removing drag assets improved overall Sharpe

Risk Management

- Max drawdown of **-4.4%** vs. S&P **-17.0%**
- Dynamic constraints tighten in crisis and loosen in calm
- Per-instrument weight caps cut noisy drag assets
- Higher returns with roughly half the volatility of the S&P

Next Steps

- Expand to intraday signals for faster event reaction
- Add an options overlay for tail-risk hedging
- Integrate live news feeds for real-time NLP scoring
- Scale the universe to international macro instruments



Thank You – Questions